

# CASE STUDY 6

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## 1. Train Search System

### Question

A user can search trains only if:

1. From Station is selected.
2. To Station is selected.
3. Journey date is selected.
4. Class is selected.

If all conditions are satisfied, train details will be displayed.

If any one condition fails, train details should be rejected.

Create a Decision Table for the above scenario.

| Requirement Number      | 1  | 2 | 3 | 4 | 5 |
|-------------------------|----|---|---|---|---|
| <u>Condition</u>        |    |   |   |   |   |
| From station selected   | F  | T | T | T | T |
| To station selected     | -- | F | T | T | T |
| Journey date selected   | -  | - | F | T | T |
| Class is selected       | -  | - | - | F | T |
| <u>Action</u>           |    |   |   |   |   |
| Train details displayed |    |   |   |   | T |
| Error message displayed | F  | F | F | F |   |

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## 2. Online Shopping System

### Question

A customer can place an order only under these conditions:

1. User is logged in.
2. Product is in stock.
3. Delivery address is available.
4. Payment is successful.

If all conditions are satisfied, order is be placed successfully.

If any condition is false, order is rejected.

Prepare a Decision Table for this banking system.

| Requirement Number         | 1  | 2 | 3 | 4 | 5 |
|----------------------------|----|---|---|---|---|
| <u>Condition</u>           |    |   |   |   |   |
| User is logged in          | F  | T | T | T | T |
| Product is in stock        | -- | F | T | T | T |
| Delivery address available | -  | - | F | T | T |
| Payment Successful         | -  | - | - | F | T |
| <u>Action</u>              |    |   |   |   |   |
| Order Placed               |    |   |   |   | T |
| Order Rejected             | F  | F | F | F |   |

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### 3. ATM Cash Withdrawal System

#### Question

A customer can withdraw cash only if:

1. ATM Card is valid.
2. Correct PIN is entered.
3. Sufficient Balance is available.
4. ATM has cash available.

If all conditions are satisfied, cash will be dispensed.

If any condition is false, the transaction fails.

Create a Decision Table for the above conditions.

| Requirement Number  | 1  | 2 | 3 | 4 | 5 |
|---------------------|----|---|---|---|---|
| <u>Condition</u>    |    |   |   |   |   |
| Card Valid          | F  | T | T | T | T |
| Correct Pin Entered | -- | F | T | T | T |
| Sufficient Balance  | -  | - | F | T | T |

|                    |   |   |   |   |   |
|--------------------|---|---|---|---|---|
| ATM Cash Available | - | - | - | F | T |
| <u>Action</u>      |   |   |   |   |   |
| Cash dispensed     |   |   |   |   | T |
| Transaction failed | F | F | F | F |   |

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## 4. Hospital Appointment Booking

### Question

A patient can book an appointment only if:

1. Patient is registered.
2. Doctor is available.
3. Appointment slot is available.
4. Payment is completed.

If all conditions are satisfied, appointment will be confirmed.

If any condition fails, booking will fail.

Prepare a Decision Table for this system.

| Requirement Number    | 1  | 2 | 3 | 4 | 5 |
|-----------------------|----|---|---|---|---|
| <u>Condition</u>      |    |   |   |   |   |
| Patient registered    | F  | T | T | T | T |
| Doctor is available   | -- | F | T | T | T |
| Slot available        | -  | - | F | T | T |
| Payment completed     | -  | - | - | F | T |
| <u>Action</u>         |    |   |   |   |   |
| Appointment confirmed |    |   |   |   | T |
| Appointment rejected  | F  | F | F | F |   |

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## 5. Student Course Registration System

### Question

A student can register for a course only if:

1. Student is enrolled.
2. Course seats are available.

3. Prerequisite course is completed.
4. Registration fee is paid.

If all conditions are true, registration will be successful.  
Otherwise, registration will fail.

Create a Decision Table for the above scenario.

| Requirement Number         | 1  | 2 | 3 | 4 | 5 |
|----------------------------|----|---|---|---|---|
| <u>Condition</u>           |    |   |   |   |   |
| Students enrolled          | F  | T | T | T | T |
| Course seats are available | -- | F | T | T | T |
| Prerequisite completed     | -  | - | F | T | T |
| Fee Paid                   | -  | - | - | F | T |
| <u>Action</u>              |    |   |   |   |   |
| Registration Successful    |    |   |   |   | T |
| Registration Failed        | F  | F | F | F |   |

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## 6. Movie Ticket Booking System

### Question

A movie ticket can be booked only if:

1. User is logged in.
2. Seats are available
3. Show timing is available
4. Payment is successful

If all conditions are satisfied, ticket will be booked.  
If any condition fail, booking will be failed.

Create a Decision Table for the above scenario.

| Requirement Number | 1  | 2 | 3 | 4 | 5 |
|--------------------|----|---|---|---|---|
| <u>Condition</u>   |    |   |   |   |   |
| User logged in     | F  | T | T | T | T |
| Seats available    | -- | F | T | T | T |
| Show available     | -  | - | F | T | T |
| Payment successful | -  | - | - | F | T |
| <u>Action</u>      |    |   |   |   |   |
| Ticket Booked      |    |   |   |   | T |
| Booking Failed     | F  | F | F | F |   |



